

1. Grey Level Transformation

Grey level transformation changes the intensity values of pixels to improve image quality, making objects more visible & enhancing contrast.

$$S = 255 - r$$

r = original pixel value

S = transformed pixel value

Given,

$$r = [50, 100, 150, 200]$$

$$S = 255 - r$$

<u>Pixel (r)</u>	<u>calculation</u>	<u>Output (s)</u>
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Transformed Pixels: [205, 155, 105, 55]

2. Log Transformation

Log transformation expands dark pixel values while compressing brighter values, making dark areas more visible.

$$S = c \log(1+r)$$

Given,

$$C = 10$$

$$\text{for } r=0$$

$$S = 10 \ln(1) = 10(0) = 0$$

$$\text{for } r=10$$

$$S = 10 \ln(11) = 10(2.3979) = 23.98$$

$$\text{for } r=50$$

$$S = 10 \ln(51) = 10(3.9318) = 39.32$$

$$\text{for } r=100$$

$$S = 10 \ln(101) = 10(4.6151) = 46.15$$

Lower intensity values increase rapidly compared to higher values, improving visibility in dark regions.

3. Histogram Processing

Histogram equalization improves image contrast by redistributing pixel intensities more uniformly.

1. Compute histogram
2. Find cumulative distribution (CDF)

$$S = (L-1) \times \text{CDF}$$

for 3-bit image

$$L = 8$$

maximum intensity = 7

$$\text{Total pixels (N)} = 2 + 2 + 4 + 8 = 16$$

<u>Intensity</u>	<u>CDF</u>	<u>New Value</u>
0	0.125	1
1	0.25	2
2	0.50	4
3	1.00	7

4. Spatial Filtering

Smoothing reduces image noise & small intensity variations before further processing.

$$\text{output} = \frac{\text{Sum of all pixels}}{9}$$

matrix:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Sum:

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

$$\text{Average: } 270 / 9 = 30$$

The Average filter smooths intensity variations & reduces random noise.

5. Spatial Filtering (Sharpening)

Sharpening emphasizes edges & fine details by enhancing intensity changes.

Use the Laplacian mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Input:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

$$\begin{aligned} &= (0 \times 10) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (4 \times 20) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (0 \times 10) \\ &= 0 - 10 + 0 - 10 + 80 - 10 + 0 - 10 + 0 = 40 \end{aligned}$$

6. Frequency Domain Filtering

(a) Differentiate low-pass & high-pass filtering

Low-pass

1. Allows low-freq components to pass
2. Removes noise & smooths the image.
3. Produces a blurred image.

High-pass

1. Allows high-freq components to pass.
2. Enhances edges & fine details
3. Produces a sharper image.

(b) Apply a low-pass filter to the frequency components

Given,

$$F = [100, 80, 40, 20]$$

A low-pass filter retains only the first two frequency components.

Filtered output:

$$[100, 80, 0, 0]$$

The high-frequency components are removed, reducing noise & producing a smoother image.

7. Thresholding

(a) Explain global thresholding in Segmentation.

Global thresholding is an image segmentation technique that separates the foreground from the background using a single threshold value. Pixels with intensity greater than or equal to the threshold are assigned 1.

(b) Apply threshold $T=100$.

Given: [200]

Rule:

- Pixel $\geq 100 \rightarrow 1$
- Pixel $< 100 \rightarrow 0$

$$200 > 100 \Rightarrow [1]$$

The pixel is classified as the foreground because its intensity is greater than threshold.

8. Edge Detection

(a) Explain the concept of edge Detection

Edge detection identifies sudden changes in pixel intensity within an image. It is used to detect object boundaries, shapes, & important image features for segmentation.

(b) Apply the Sobel horizontal operator.

Sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image: $\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$

$$(-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -10 - 20 - 10 = -40$$

$$0$$

$$(1 \times 30) + (2 \times 30) + (1 \times 30) = 30 + 60 + 30 = 120$$

$$120 - 40 = 80$$

$$\boxed{\text{Answer} = 80}$$

9. Region Based Segmentation

a) Region growing is a segmentation method that starts from a selected seed pixel & groups neighbouring pixels whose intensity values are similar to the seed within a specified threshold.

b) Given, [48, 50, 52]

Seed value: 50

Threshold difference: 5

<u>Pixel</u>	<u>Difference</u>
48	2
50	0
52	2

Answer: [48, 50, 52]

10. Combined Enhancement & Segmentation

(a) Smoothing is applied before segmentation to reduce image noise & unwanted intensity variations. This improves segmentation accuracy by preventing false boundaries.

(b) Given, 48

Threshold: $T = 50$

Rule:

• Pixel $\geq 50 \rightarrow 1$

• Pixel $< 50 \rightarrow 0$

Since $48 < 50$

Segmented output: 0

The smoothed pixel value is below the threshold.

11. Power-Law (Gamma) Transformation

(a) Gamma correction adjusts the brightness of an image by modifying pixel intensity values using a power-law function.

(b) $S = Cr^\gamma$, $C=1$ & $\gamma=0.5$

Since $\gamma=0.5 < 1$, the transformation increases the brightness of low-intensity pixels.

$$S = Cr^\gamma$$

$$\gamma^{0.5} = \sqrt{\gamma}$$

$$C=1, \gamma=0.5$$

<u>Pixel (r)</u>	<u>calculation</u>	<u>output (s)</u>
25	$\sqrt{25}$	5
100	$\sqrt{100}$	10
225	$\sqrt{225}$	15

Answer: [5, 10, 15]

12. Contrast Stretching

(a) Contrast stretching improves image quality by expanding a narrow range of intensity values to the full grayscale range.

(b) linear transformation to range [0, 255]

$$S = \frac{(r - r_{\min})}{(r_{\max} - r_{\min})} \times 255$$

$$r_{\min}=50, r_{\max}=150$$

$$r=50$$

$$S = \frac{(50-50)}{100} \times 255 = 0$$

$$r=100$$

$$S = \frac{100-50}{100} \times 255 = 127.5 \approx 128$$

$$r=150$$

$$S = \frac{150-50}{100} \times 255 = 255$$

Answer: [0, 128, 255]

13. Histogram Equalization

Given,

$$\text{Total Pixels (N)} = 4 + 6 + 10 + 6 = 26$$

<u>Intensity</u>	<u>Frequency</u>	<u>CDF</u>
0	4	$4/26 = 0.154$
1	6	$10/26 = 0.385$
2	10	$20/26 = 0.769$
3	6	$26/26 = 1.000$

for 4-bit image, $L = 16$

$$L-1 = 15$$

$$S = 15 \times \text{CDF}$$

<u>Intensity</u>	<u>New Intensity</u>
0	2
1	6
2	12
3	15

14. Median Filtering

(a) Salt & Pepper noise appears as randomly occurring black & white pixels in an image.

Given,

$$\begin{bmatrix} 10 & 200 & 10 \\ 200 & 50 & 200 \\ 10 & 200 & 10 \end{bmatrix}$$

Arrange values in ascending order:

$$10, 10, 10, 10, 50, 200, 200, 200, 200$$

median (5th value): 50

The median filter removes impulse noise while maintaining important image edges, producing a cleaner image.

15.

Gaussian Smoothing

- (a) Gaussian smoothing reduces image noise & small intensity variations while preserving important image structures better than simple averaging.
- (b) Justify Gaussian filter assigns higher weights to the center pixels & lower weights to neighbouring pixels.

Kernel:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image: $\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$

Weighted Sum:

$$\begin{aligned} & (1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (4 \times 40) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (1 \times 10) \\ & = 10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10 \\ & = 360 \end{aligned}$$

Sum of kernel weights:

$$1 + 2 + 1 + 2 + 4 + 2 + 1 + 2 + 1 = 16$$

Center pixel:

$$\frac{360}{16} = 22.5$$

output: $\boxed{22.5}$